

Network and Transport Layer

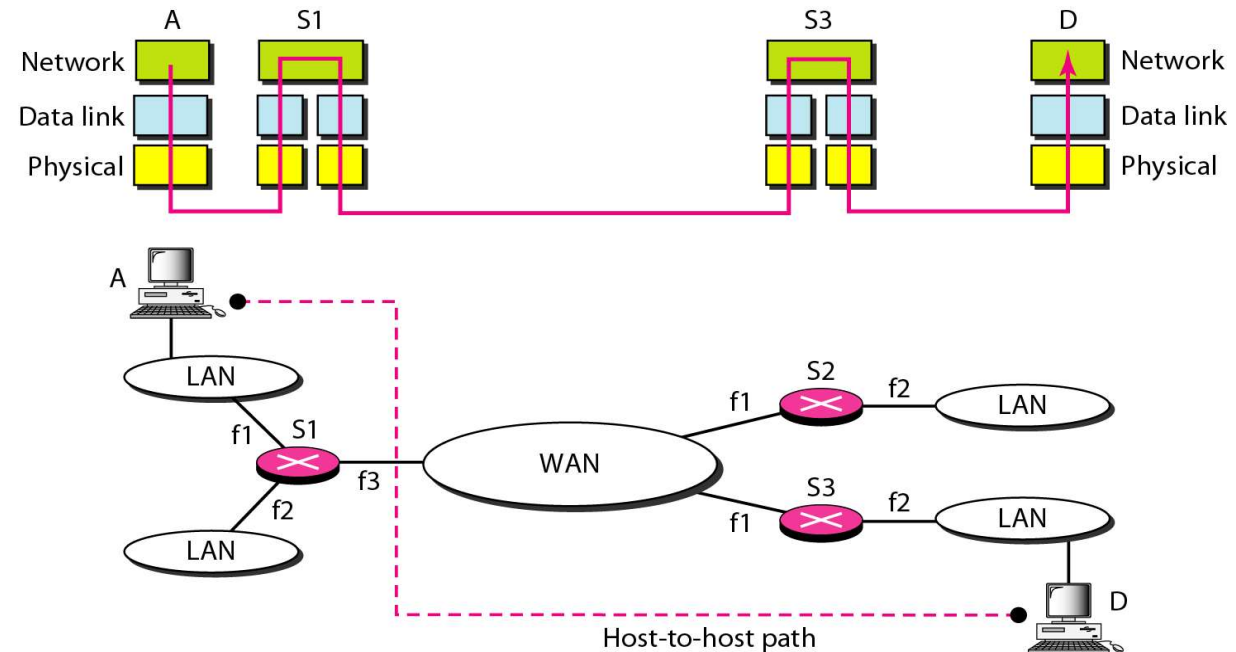
Unit-4: Network and Transport Layer:

Host to Host Delivery: Internetworking, Addressing and Routing, Network Layer Protocols: ARP, IPV4, ICMP, IPV6

Transport Layer: Process to Process Delivery: UDP, TCP, Congestion Control. Routing Protocols: AODV, DSDV, DSR.

Internetworking

- The communication at the network layer is **host-to-host**.
- Internetworking is the process of connecting different networks by using intermediary devices such as **routers or gateway** devices.
- The interconnected networks are called an **internetwork**, or simply an **internet**.
- Internetworking is **only possible** when **all the connected networks use the same protocol stack or communication methodologies**.



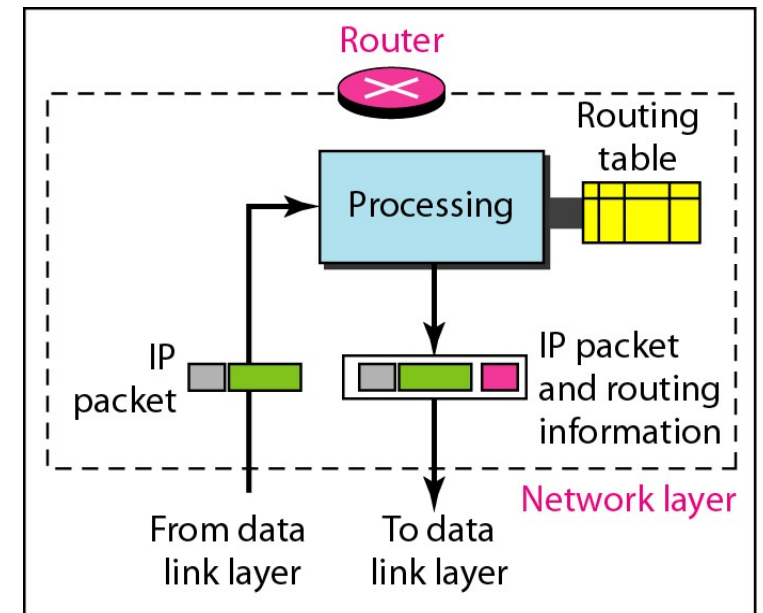
Router

Router:

- ❖ A **router** is a **three-layer device**; physical, data link and network layer.
- ❖ As a physical layer device it can regenerate bit, as data link layer device the router can check the physical addresses contained in a packet and as a network layer device it can check the network layer addresses (addresses in IP layer).

Routing:

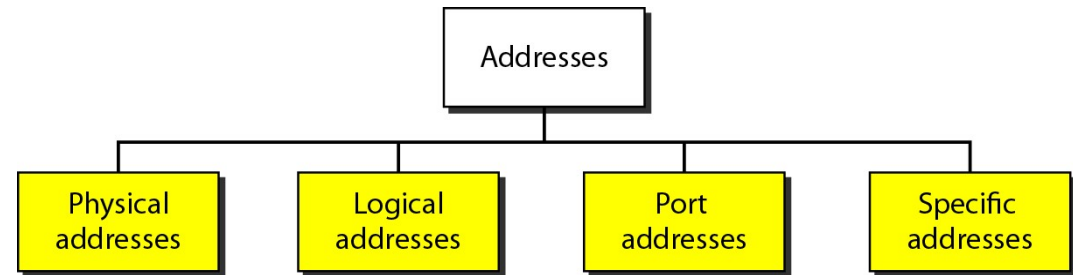
- ❖ **Routing** refers to the process of selecting the **shortest** and the most **reliable path** over which to send data to its ultimate destination.
- ❖ IP routing protocol makes the distinction between hosts and gateways.
- ❖ A host is the end system to which data is ultimately deliverable.
- ❖ An IP gateway, on the other hand, is the router that accomplishes the act of routing data between two networks.



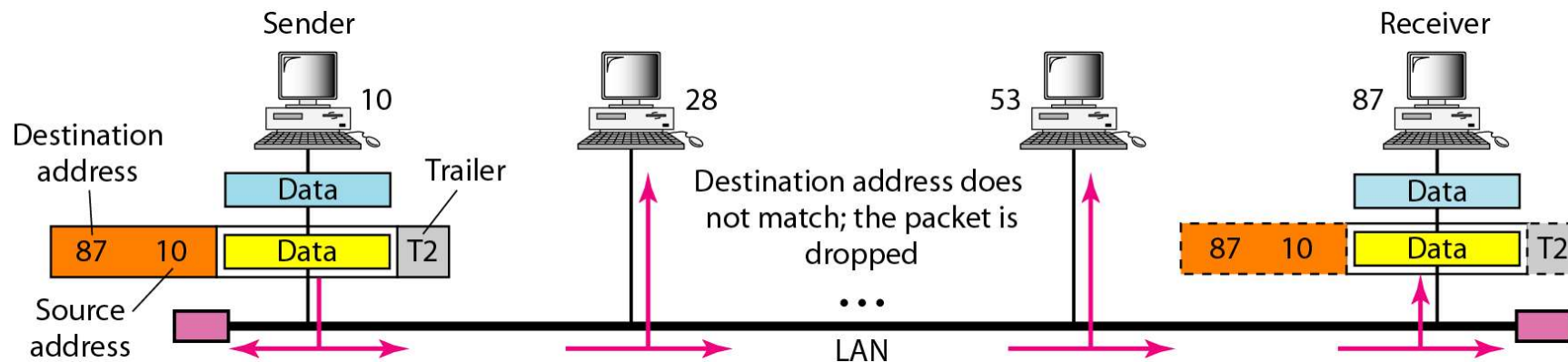
c. Network layer at a router

Examples: Physical Address

Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.



In Figure, a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link (bus topology LAN). As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.



Physical address

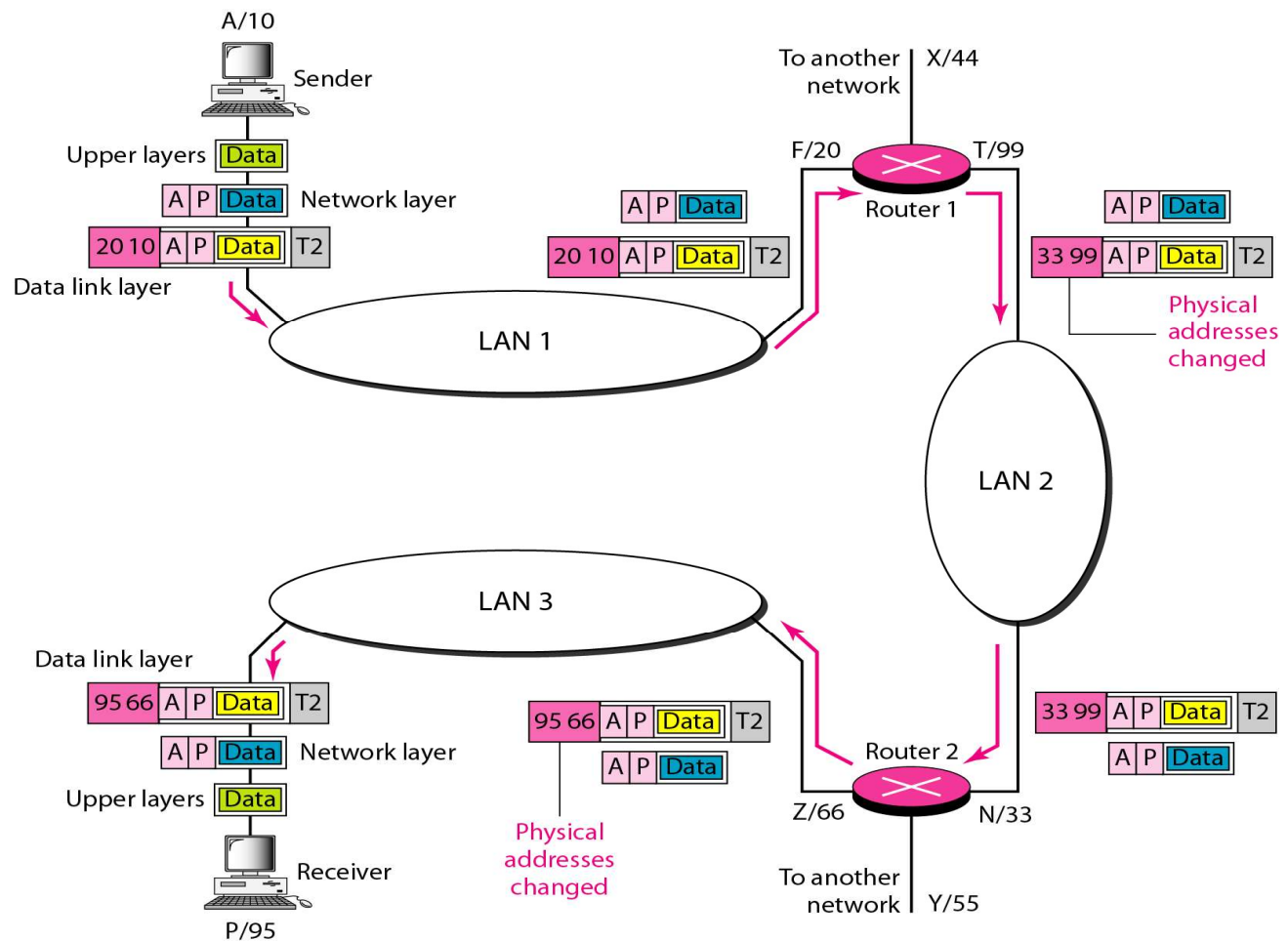
Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:

07:01:02:01:2C:4B

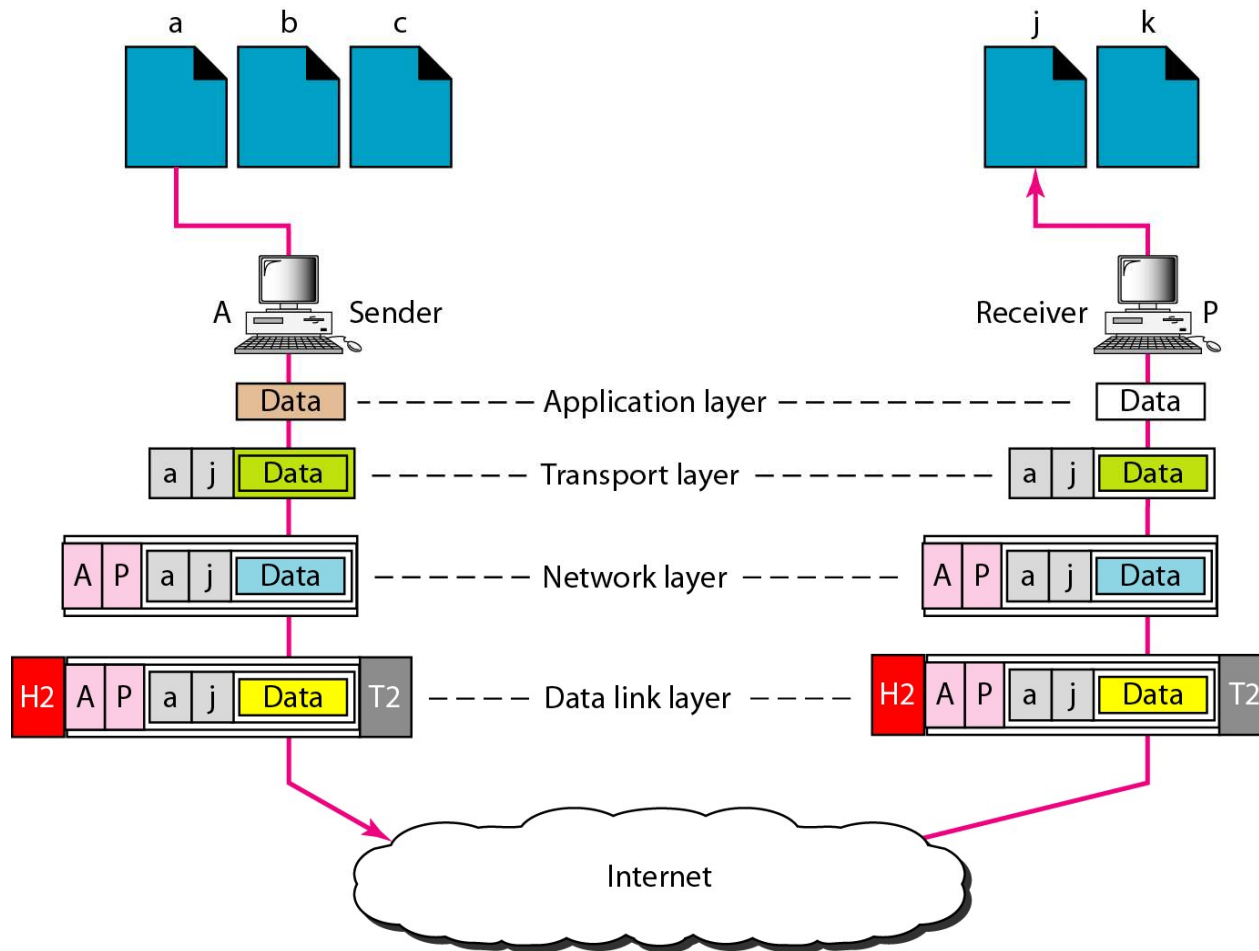
A 6-byte (12 hexadecimal digits) physical address.

Examples: Physical & Logical addresses

Figure shows a part of an internet with two routers connecting three LANs. Each device (computer or router) has a pair of addresses (logical and physical) for each connection. Each router is connected to three networks (only two are shown in the figure). So each router has three pairs of addresses, one for each connection.



Port addresses



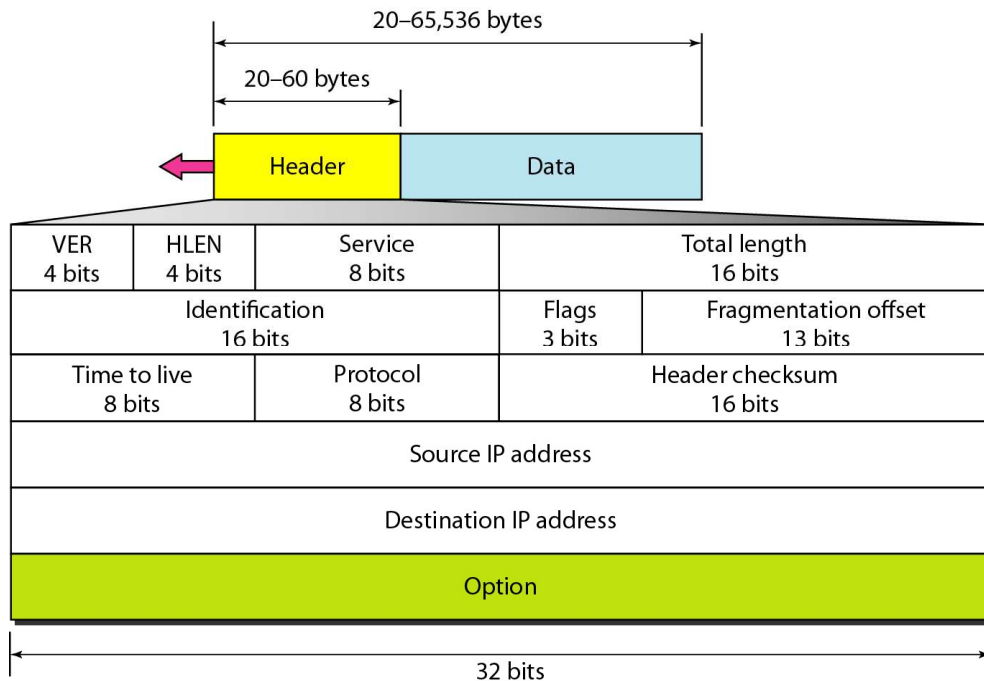
Port addresses

A port address is a 16-bit address represented by one decimal number as shown.

753

**A 16-bit port address represented
as one single number.**

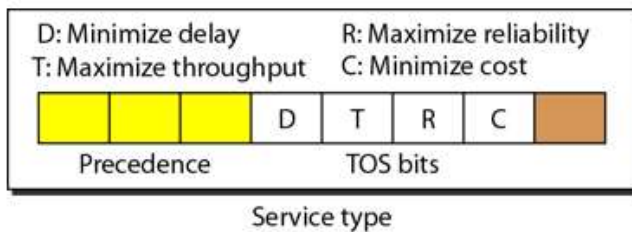
IPv4 Protocol



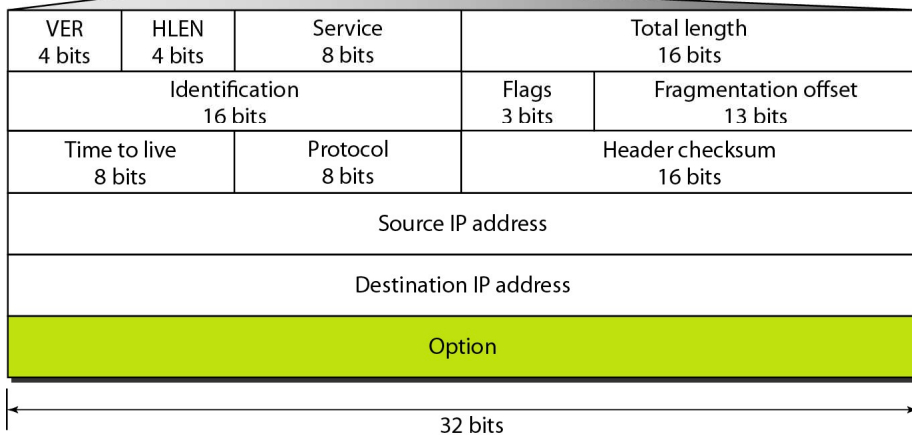
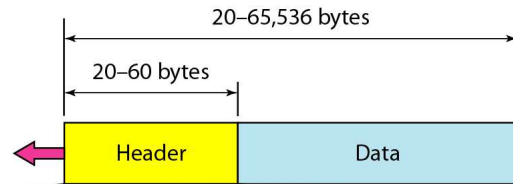
Version	This is a 4-bit field, and it tells the version of IP we are using.
HLen	It gives the length of the header. The minimum header length should be 20 bytes, as the maximum header size of 4 bits is 15. If we use the options field, 60 Bytes(20+40) will be the maximum header length.
Type of Service	The first 3 bits tell the precedence, and the next 4 bits tell the type of service, and the last bits are not used. The 4 bits of services define the delay, throughput, reliability, cost.
Total Length	The total length field defines the total length of the datagram, including the header. total length can be calculated as data length + header length or data length = total length – header length of 16 bits

Table Types of service

TOS Bits	Description
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay



IPv4 Protocol



Protocol field and encapsulated data

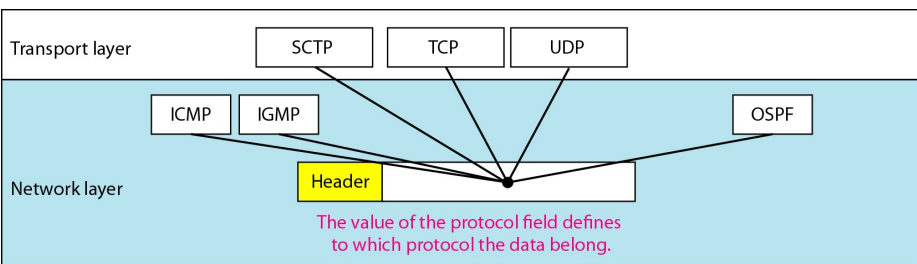
Identification (Fragment ID)	As IPv4 is a datagram service, these bits help to identify uniquely when fragmentation is over on datagram packets.
Flags	The flag indicates the fragmentation using the identification field, basically tells whether it can be fragmented or the current one is the final one.
Fragment Offset	The neighbour Position of every fragment is measured from the original data's beginning by units of 8 bytes.
Time to Live	This helps to misguide the transactions of datagrams. It measures the no.of.routers that can datagram pass through. It checks by decrementing the value to 1 till it reaches 0. Datagram gets discarded when it reaches zero.
Protocol	IPv4 contains data from various protocol. This field helps the network layer to understand which data belongs to which protocol.
Header Checksum	This field is used to detect errors in the packets or messages.
Source IP Address	32-bit address of the sending host.
Destination Ip Address	32-bit address of the receiving host.
Options	Every datagram doesn't specify the option field. It is the list of specifications that contains security restrictions, routing, etc.

Flags used in fragmentation



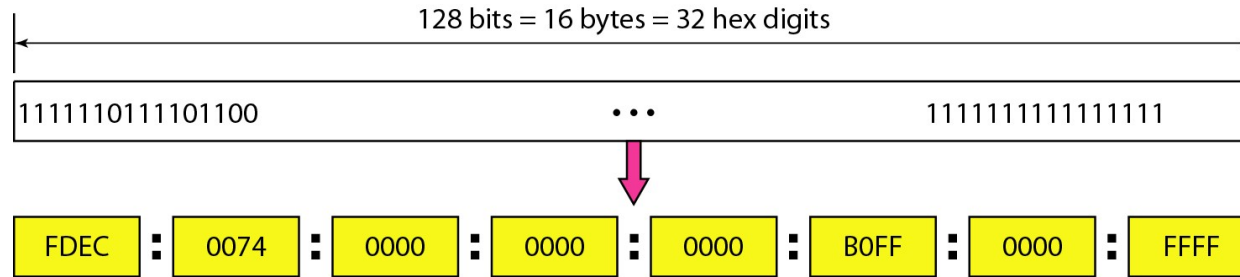
D: Do not fragment
M: More fragments

- bit 0: Reserved (always 0)
- bit 1 (DF - Don't Fragment): If 1 → packet cannot be fragmented.
- bit 2 (MF - More Fragments): If 1 → more fragments follow; if 0 → last fragment.

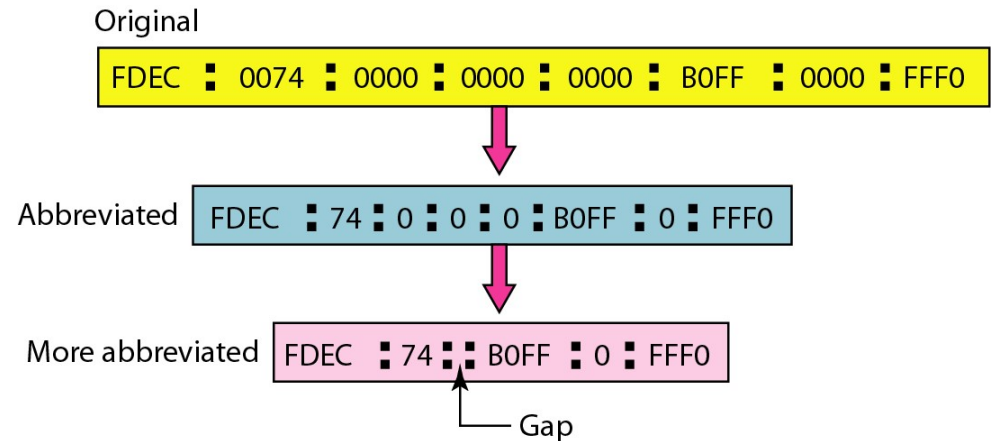


IPv6 Protocol

- The Internet Protocol version 6 (IPv6) is the future addressing format.
- In recent time it has been noticed that the IPv4 addresses getting exhausted with a rising number of machines and the public IPs being structured up will go into version 6. An IPv6 address is 128 bits.



- **Length:** 128 bits (compared to 32 bits in IPv4)
- **Representation:** Written as **8 groups of 16 bits** each
- Each 16-bit is represented as **4 hexadecimal digits**
- Groups are separated by **colons (:)**



IPv6 Protocol

Version (4-bits)

It shows the version of the internet protocol we used, i.e. 0110

Traffic Class (8-bits)

This is an 8-bit field in which 8 bits are divided into **two** parts. The **most significant 6-bit** is for the type of **service** so that the router will get to know about what services need to be provided to the given packet. And for **Explicit Congestion Notification** (ECN), the **least significant 2-bit** is used.

Flow Label (20-bits)

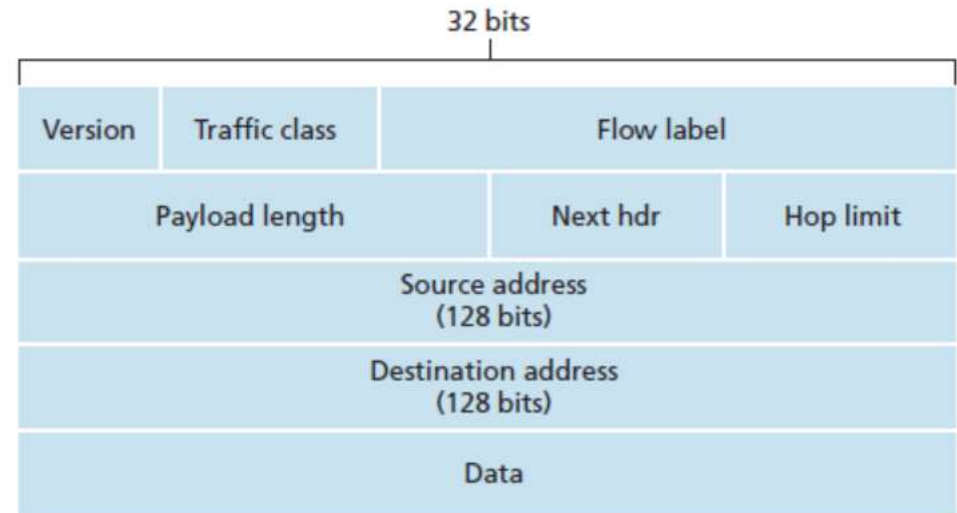
This 20-bit is required for maintaining the sequential flow of packets related to a particular communication.

Payload Length (16-bits)

This field is used to help the router know how much information is stored in the payload of a particular packet.

Next Header (8-bits)

This field is used to represent the type of extension header



Hop Limit (8-bits)

Hop limit is a field in a header that **stops** the header from going into an **infinite loop** in the network.

Source Address (128-bits)

This field provides the address from where the packet originates.

Destination Address (128-bits)

The destination address is the address of the packet's intended recipient.

IPv6 vs IPv4 Protocol

Enhanced Security

IPv4 didn't have any built-in security features. But, IPv6 comes with built in support for [IPsec](#), enhancing privacy, data integrity, and secure authentication across the internet.

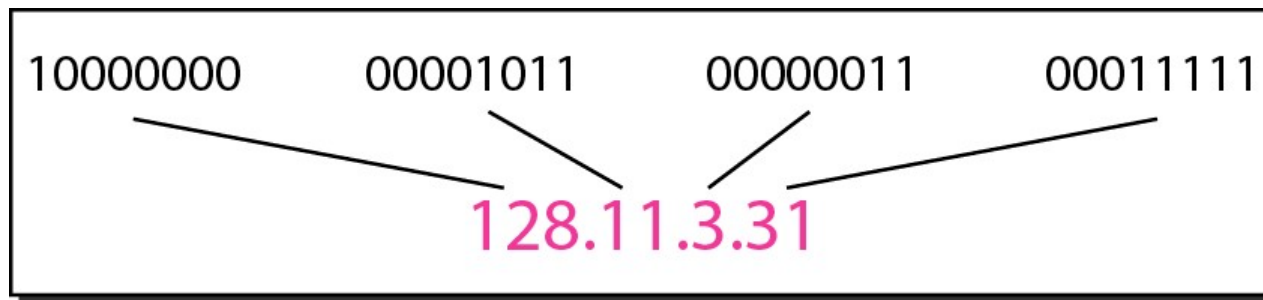
Improved Network Efficiency and Performance

- Simplified packet headers for more efficient routing and processing.
- Eliminates the need for NAT (Network Address Translation), which can complicate communications and reduce performance.

Feature	IPv4	IPv6
Address Size	32 bits	128 bits
Address Format	Dotted decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
Address Space	About 4.3 billion addresses	About 340 undecillion addresses
NAT	Required due to address shortage	Not needed because of vast address space
IPSec Support	Optional	Built-in
Autoconfiguration	Manual setup or DHCP	Automatic address configuration
Header Complexity	Complex	Simple
Fragmentation	Handled by routers and sending hosts	Handled by sending hosts only

IPv4 Addresses

- An *IPv4 address* is a *32-bit* address that uniquely and universally defines the connection of a device
- The address space of IPv4 is 2^{32} or 4,294,967,296.



IPv4 Addresses

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

IPv4 Addresses

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. *There must be no leading zero (045).*
- b. *There can be no more than four numbers.*
- c. *Each number needs to be less than or equal to 255.*
- d. *A mixture of binary notation and dotted-decimal notation is not allowed.*

Classful Addressing

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

An IP address is divided into two parts:

- **Network ID (Netid):** It represents the number of networks.
- **Host ID (Hostid):** It represents the number of hosts.

Netid and Hostid

The **netid** is in color, the hostid is in white.

Example : In class A, first byte defines the netid and last three bytes define the hostid.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Default Masks for Classful addressing

In an **IP address**, a **mask** (also called a **subnet mask**) is used to **separate the network part** and the **host part** of the address.

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Class	Left-most bit	Starting IP address	Last IP address
A	0xxx	0.0.0.0	127.255.255.255
B	10xx	128.0.0.0	191.255.255.255
C	110x	192.0.0.0	223.255.255.255
D	1110	224.0.0.0	239.255.255.255
E	1111	240.0.0.0	255.255.255.255

Classful Addressing

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Classless Addressing

- ❖ In **classful addressing**, a large part of the available addresses were wasted.
- ❖ Classful addressing, which is almost obsolete, is replaced with classless addressing.
- ❖ In **classless addressing**, the network and host portions are **not fixed**. We use **Classless Inter-Domain Routing CIDR notation (/n)** to specify how many bits belong to the network.
- ❖ In IPv4 addressing, a block of addresses can be defined as x.y.z.t /n in which x.y.z.t defines one of the addresses and the /n defines the mask.

Feature	Classful	Classless
Network ID Size	Fixed (based on class)	Flexible (user-defined)
Subnet Mask	Default (e.g., /8, /16, /24)	Custom (e.g., /20, /26)
Routing Protocol	Does not send mask info	Sends mask info
IP Utilization	Inefficient	Efficient
Supports VLSM	 No	 Yes
Example	192.168.1.0 (Class C)	192.168.1.0/26

Classless Addressing

- ✓ Each address in the block can be considered as a two-level hierarchical structure.
- ✓ Prefix, defines the **network** (n bits). Suffix, defines **connection of a device** to the Internet ($32-n$ bits)
- ✓ The **first address** in the block can be found by setting the rightmost $(32 - n)$ bits to 0s.
- ✓ The **last address** in the block can be found by setting the rightmost $(32 - n)$ bits to 1s.
- ✓ The **number of addresses** in the block 2^{32-n}

Network address	The first address in a block of IP addresses
Broadcast address	The last address in a block of IP addresses



Numerical

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

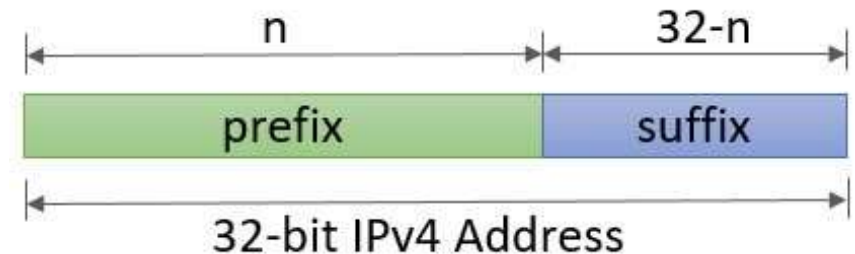
11001101 00010000 00100101 0010 0111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 0001 0000

or

205.16.37.32.



Subnetting

- ✓ Subnetting is a method of dividing a single physical network into logical sub-networks (subnets).
- ✓ It improves **network efficiency, security, and management**.
- ✓ In **FLSM**, all subnets have the **same subnet mask** and therefore **equal number of hosts**.
- ✓ In **VLSM**, subnets can have **different subnet masks**, allowing for **different host counts** per subnet. It's more **efficient** because it avoids IP wastage.

Attributes	FLSM	VLSM
Sub-net size	Equal	Variable
Sub-net mask	Same	Different
Number of Hosts	Equal	Variable
Configuration	Easy and simple	Complex
IP addresses wastage	More	Less
Efficiency	Less	More
Routing Protocols	Supports classful and classless	Supports classless
Applications	Suitable for private IP addresses	Suitable for public IP addresses

Numerical

Problem 1

Number of needed subnets **14**

Number of needed usable hosts **14**

Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets -	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 .	0	0	0	0	0	0	0	0		

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
+16
240

16	Observe the total number of hosts.
-2	
14	Subtract 2 for the number of usable hosts.

Numerical

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	165	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 16 \\
 8 \\
 4 \\
 2 \\
 +1 \\
 \hline
 255
 \end{array}$$

$$\begin{array}{r}
 64 \\
 -2 \\
 \hline
 62
 \end{array}$$

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

Numerical

Custom Subnet Masks

Problem 3

Network Address **148.75.0.0 /26**

/26 indicates the total number of bits used for the network and subnetwork portion of the address. All bits remaining belong to the host portion of the address.

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Numerical

The given IP addresses is 201.20.30.40/24. Find out the (i) subnet mask, (ii) Network ID, (iii) 4th host ID, (iv) last host ID, (v) broadcast address, (vi) number of subnet, (vii) valid hosts

Number of bits borrowed = $24 - 24 = 0$, Number of Network = $2^0 = 1$, Number of hosts = 256, Usable = 254,

- i. Default subnet mask = 255.255.255.0 (/24)
- ii. Network ID = 201.20.30.0
- iii. 4th host ID = 201.20.30.4
- iv. Last host ID (last usable) = 201.20.30.254
- v. Broadcast address = 201.20.30.255
- vi. Number of subnets = 1 (using default /24, no further subnetting)
- vii. Valid hosts per subnet = $2^8 - 2$

Numerical

The given IP addresses is 183.42.0.0/22. Find out (i) the Class, (ii) default subnet mask, (iii) Custom mask, (iv) number of networks (v) number of hosts per subnet (vi) valid subnets (vii) Broadcast address of each subnet?

- **Borrowed bits** = $22 - 16 = 6$ bits
- **Number of subnets (Networks)**
= $2^6 = 64$ subnets
- **Host bits** = $32 - 22 = 10$ bits
- **Number of host** = $2^{10} = 1024$
- **Number of usable hosts** = $1024 - 2$
= 1022 hosts / subnet

Parameter	Value
IP Address	183.42.0.0/22
Class	B
Default Mask	255.255.0.0 (/16)
Custom Mask	255.255.252.0 (/22)
No. of Subnets	64
Hosts per Subnet	1022
Valid Subnets	183.42.0.0, 183.42.4.0, 183.42.8.0, ...
Broadcasts	183.42.3.255, 183.42.7.255, 183.42.11.255, ...